Which model does best for Sirius

Conclusions, and the recommended model in precise terms

Vera Schroeder's bimanual interference data · prepared for Krešimir Josić · 3 November 2026

This note states the conclusions of the full comparison (28 models, 4 chains × 5000 draws each; see the longer report, *Sirius_model_comparison_5000_draws.pdf*) and writes the recommended model, Model E, exactly as it was fitted. Scores are expected log predictive densities (elpd, log-likelihood units summed over trials; higher is better); "±" gives one standard error.

1 Conclusions

1.1 Which model does best

No single model wins. Five models are tied at the top (Table 1): their scores span 2.6 units at the trial level and 2.2 units for new sessions, less than their standard errors of 2–5. All five share the same structure:

(i) each session has its own bias, slope and lapse rates;

(ii) a distractor shifts the psychometric curve by a fixed amount plus an amount proportional to its amplitude, and bimanual trials have their own lapse rates, which change with the distractor amplitude;

(iii) the session bias follows a trend across the study.

We recommend Model E because it is the simplest of the five: its distractor terms are the same in every session, so it has 12 parameters per session (E+shared 13, the three D-based models 16) and 123 effective parameters (the D-based models 144), while predicting as well as the others. The next models, D+shared and D3+crit, are 7.9 and 8.5 units behind E for new sessions; all others are further behind.

1.2 What Model E says

- **A distractor makes the target feel stronger.** On bimanual trials the PSE shifts by $-4.0\,\mu m$ for the left hand (95% interval -4.7 to -3.3) and $-5.9\,\mu m$ for the right (-6.7 to -5.3), averaged over sessions, so a weaker target is enough for a "high" answer. The shift grows with the distractor: from -1.4 (left) and $-3.6\,\mu m$ (right) in the $6\,\mu m$ sessions to -6.1 and $-7.7\,\mu m$ in the $32\,\mu m$ sessions.
- **The two hands differ in the form of the shift.** The amplitude term is similar for the two hands ($\beta_{2,L} = 0.72$, $\beta_{2,R} = 0.66$ per unit of d). For the left hand the shift is proportional to the distractor amplitude: the presence term $\beta_{3,L} = 0.15$ (-0.40 to 0.66) is indistinguishable from zero. The right hand has an extra shift for any distractor, $\beta_{3,R} = 0.76$ (0.24 to 1.29).
- **Sensitivity does not change.** E has one slope for both conditions; adding a separate bimanual slope to Model D does not improve prediction (1.5 ± 3.4 at the trial level, 0.9 ± 2.8 for new sessions). In E the JND is slightly larger on bimanual trials (by $0.35\,\mu m$ left, $0.23\,\mu m$ right) only because of the bimanual lapses; Model B, with its separate bimanual slopes, gives $1.03\,\mu m$ for the left hand.

Table 1. The five tied models (top) and the next models and references (bottom), each minus Model E (positive: better than E). "Trial-level": PSIS-LOO over trials, as in Vera's reports. "New sessions": leave-one-session-out. Standard errors are computed from the 42 per-session differences. p_{loo}: effective number of parameters. D is Vera's Model C with separate, amplitude-dependent bimanual lapse rates and a distractor-presence term; "trend": the session bias follows the session's distractor amplitude (or time, in D+order). Definitions in Section 3.

Model	What it is	Parameters per session	p_{loo}	Trial-level minus E (SE)	New sessions minus E (SE)
E	D with fixed distractor terms + trend	12	123	0	0
E+shared	E + daily bias shared by both hands	13	121	0.6 (1.8)	2.2 (2.8)
D+crit	D + trend with distractor amplitude	16	144	2.6 (4.0)	1.6 (2.9)
D+order	D + trend linear in time	16	144	2.0 (4.2)	1.1 (5.1)
D+crit+order	D + both trends	16	144	2.5 (4.0)	2.0 (3.7)
D+shared	D + daily bias shared by both hands	17	141	1.8 (4.5)	−7.9 (4.8)
D3+crit	E, only the bias varies by session	2	84	−14.0 (6.8)	−8.5 (4.5)
D	previous recommendation (no trend)	16	145	−1.6 (4.5)	−14.6 (5.6)
D4	E without the trend	12	125	−4.3 (2.0)	−15.7 (4.6)
B	Vera's Model B	16	179	−14.4 (9.8)	−34.2 (10.6)
C	Vera's Model C	10	139	−51.2 (13.3)	−70.5 (13.8)

- **Distractors cause extra "high" answers.** In a typical session lapses are rare on unimanual trials: 0.1% "high" and 0.6% "low" for the left hand, 0.2% and 0.6% for the right. On bimanual trials the left hand's "high" lapse rate rises from 0.3% (0.0 to 0.8%) with a 6 µm distractor to 8.9% (4.7 to 13.4%) with a 32 µm one. The right hand's is raised at every amplitude (2.3% at 6 µm, 5.3% at 32 µm); its rise with amplitude is not clear ($\kappa_R^{\text{hi}} = 0.64$, -0.27 to 1.66).
- **The criterion moved.** In a typical session the unimanual PSE is 2.9 µm (left; 95% interval 0.9 to 4.9) and 6.1 µm (right; 3.8 to 8.2) higher in the 32 µm sessions than in the 6 µm sessions, comparable to the distractor effect itself.

1.3 How confident we can be

Well beyond the noise.

- **The tied models beat Vera's.** For predicting a new session, E is ahead of B by 34.2 ± 10.6 and of C by 70.5 ± 13.8.
- **The criterion trend is needed.** Removing it from E (model D4) lowers the score for new sessions by 15.7 ± 4.6, with 34 of the 42 sessions predicted less well, and at the trial level by 4.3 ± 2.0.
- **Bimanual trials need their own lapse rates.** Giving them their own (C+L against C) improves prediction of new sessions by 47.0 ± 11.4. For the left hand the raw counts show the rise with the distractor directly: no weak-target ($\leq$ 12 µm) "high" answers in 262 bimanual trials with 6–12 µm distractors, but 36 in 447 with 24–32 µm distractors. The observed rates of weak-target "high" and strong-target ($\geq$ 44 µm) "low" answers on bimanual trials, in 20 cells (hand × error type × five distractor-amplitude bins), lie inside E's 95% predictive intervals in 20 cells, B's in 19 and C's in 17.

- **The computation is reliable.** No fit had divergent transitions; $\hat{R} \leq 1.004$ and at least 1735 effective draws for every parameter. Chains five times longer than in the first comparison moved the scores by at most 1.0 units. The leave-one-session-out estimates agreed with exact refits to within 0.2 and changed by at most 0.01 with a new random seed (checked for B, D, D+crit and E).

Not settled by these data.

- **Which of the five tied models is best.** Their scores differ by less than one standard error; E is preferred only because it is the simplest.
- **Why the criterion moved.** A criterion that follows the session's distractor amplitude and one that drifts linearly in time fit equally well (difference 0.5 ± 3.8 for new sessions), because the distractor amplitude was raised over the study (correlation 0.91 with session order).
- **How much the amplitude dependence of the bimanual lapses adds.** For the left hand it is clear in the parameters ($\kappa_L^{\mathrm{hi}} = 2.51$, 1.15 to 3.91) and in the raw counts, but it improves prediction of new sessions by only 5.8 ± 4.5 (D against D without it, C+LP): without it, the session-to-session variation of the lapse rates takes up part of the pattern.
- **The size of the right hand's fixed shift.** It is clearly positive in every tied model, but it fell from 1.03 (Model D) to 0.76 when the trend was added, so its size depends on how the trend is modelled.
- **Vera's trial-level score alone is not decisive**: on it, E is ahead of B by only 14.4 ± 9.8. The clear separation is in predicting new sessions.
- **The mechanism of the bimanual errors.** Replacing D's amplitude-dependent lapses by a fixed fraction of bimanual trials on which the monkey judges the distractor, or the stronger of the two vibrations, fits slightly but not clearly worse (-4.5 ± 4.4 and -4.8 ± 4.2 for new sessions). The lapse terms describe the errors; they do not identify what causes them.

Caveats. The data come from one monkey, and with 28 models compared, differences of a few units should not be trusted. The step that would raise confidence most is a model-recovery study: simulate data from E and from B, refit both, and check that the comparison picks the model that generated the data.

2 Model E in precise terms

2.1 Data and notation

Trials $i = 1, \ldots, N$ ($N = 12{,}937$); sessions $s = 1, \ldots, S$ in date order ($S = 42$, each with both hands); stimulated hand $h \in \{L, R\}$; $h(i)$ and $s(i)$ are the hand and session of trial i. The response is $y_i = 1$ if Sirius answered "high" and $y_i = 0$ otherwise; $c_i = 1$ marks bimanual trials. The target amplitude $a_i \in \{6, 12, 18, 24, 32, 38, 44, 50\}$ µm enters as

$$x_i = \frac{a_i - 28}{\sigma_a}, \qquad \sigma_a = 14.63\,\mu\mathrm{m} \text{ (the standard deviation of the eight amplitudes).}$$

Each session has one distractor amplitude a_s^{dist} (6 to 32 µm), with $d_s = a_s^{\mathrm{dist}}/\sigma_a$, $d_i = c_i\, d_{s(i)}$ and $\bar{d} = S^{-1} \sum_s d_s = 1.432$ (21.0 µm). $\mathrm{logistic}(u) = 1/(1 + e^{-u})$, and $\mathrm{HalfNormal}(\tau)$ is the distribution

of $|X|$ with $X \sim \mathcal{N}(0, \tau^2)$.

2.2 Likelihood

Given the parameters, the responses are independent with

$$y_i \sim \text{Bernoulli}(p_i), \qquad p_i = \gamma_i^{\text{hi}} + \left(1 - \gamma_i^{\text{hi}} - \gamma_i^{\text{lo}}\right) \text{logistic}(\eta_i),$$

$$\eta_i = \beta_{0,hs} + \beta_{1,hs}\, x_i + \beta_{2,h}\, d_i + \beta_{3,h}\, c_i, \qquad \gamma_i^k = \begin{cases} \frac{1}{4}\,\text{logistic}\left(\omega_{hs}^k\right) & c_i = 0, \\ \frac{1}{4}\,\text{logistic}\left(\omega_{hs}^{k,\text{bi}}\right) & c_i = 1, \end{cases}$$

with $h = h(i)$, $s = s(i)$ and $k \in \{\text{hi}, \text{lo}\}$. γ_i^{hi} and γ_i^{lo} are the probabilities of answering "high" or "low" regardless of the target; each lies in $(0, \frac{1}{4})$. (For numerical stability p_i is clipped to $[10^{-8}, 1 - 10^{-8}]$.)

2.3 Session level

For each hand h and session s, with all z's independent standard normal,

$$\beta_{0,hs} = \mu_{0,h} + \lambda_h\,(d_s - \bar{d}) + \sigma_{0,h}\, z_{hs}^{(0)}, \qquad \omega_{hs}^k = \mu_h^k + \sigma_h^k\, z_{hs}^{(k)},$$

$$\beta_{1,hs} = \mu_{1,h} + \sigma_{1,h}\, z_{hs}^{(1)}, \qquad \omega_{hs}^{k,\text{bi}} = \mu_h^{k,\text{bi}} + \kappa_h^k\,(d_s - \bar{d}) + \sigma_h^{k,\text{bi}}\, z_{hs}^{(k,\text{bi})}.$$

The distractor terms $\beta_{2,h}$ and $\beta_{3,h}$ are the same in every session. λ_h is the criterion trend: it moves the bias of both conditions with the session's distractor amplitude. κ_h^k makes the bimanual lapse rates depend on the distractor amplitude. Per session there are 6 parameters per hand ($\beta_0, \beta_1, \omega^{\text{hi}}, \omega^{\text{lo}}, \omega^{\text{hi,bi}}, \omega^{\text{lo,bi}}$), 12 in all; there are 17 hand-level parameters per hand, 34 in all.

2.4 Priors

Independently for each hand h and lapse type k:

$\mu_{0,h} \sim \mathcal{N}(0, 2^2)$	$\sigma_{0,h} \sim \text{HalfNormal}(2)$	$\lambda_h \sim \mathcal{N}(0, 1)$
$\mu_{1,h} \sim \text{LogNormal}(\log 4,\ 0.5^2)$	$\sigma_{1,h} \sim \text{Exponential}(1)$	
$\beta_{2,h} \sim \mathcal{N}(0, 1)$	$\beta_{3,h} \sim \mathcal{N}(0, 1)$	
$\mu_h^k \sim \mathcal{N}(-3, 1.5^2)$	$\sigma_h^k \sim \text{HalfNormal}(0.5)$	
$\mu_h^{k,\text{bi}} \sim \mathcal{N}(-3, 1.5^2)$	$\sigma_h^{k,\text{bi}} \sim \text{HalfNormal}(0.5)$	$\kappa_h^k \sim \mathcal{N}(0, 2^2)$

All but λ_h, $\beta_{3,h}$ and the bimanual lapse parameters are Vera's priors for Model C ($\beta_{2,h}$ takes the prior of C's mean $\mu_{2,h}$). $\mu_h^k = -3$ corresponds to a lapse rate of $\frac{1}{4}\,\text{logistic}(-3) = 1.2\%$.

2.5 Derived quantities

For hand h, session s and condition $c \in \{0, 1\}$, with $\gamma^{\text{hi}}, \gamma^{\text{lo}}$ the lapse rates of that condition:

$$\text{PSE (50\% point):} \quad a_{hsc}^* = 28 + \frac{\sigma_a}{\beta_{1,hs}}\left[\ln\frac{1 - 2\gamma^{\text{hi}}}{1 - 2\gamma^{\text{lo}}} - \beta_{0,hs} - c\,\left(\beta_{2,h}\, d_s + \beta_{3,h}\right)\right],$$

$$\text{PSE shift:} \quad \Delta_{hs} = a_{hs1}^* - a_{hs0}^*,$$

$$\text{JND (half the 25–75\% distance):} \quad J_{hsc} = \frac{\sigma_a}{2\beta_{1,hs}}\,\ln\frac{(3 - 4\gamma^{\text{hi}})(3 - 4\gamma^{\text{lo}})}{(1 - 4\gamma^{\text{hi}})(1 - 4\gamma^{\text{lo}})}.$$

The distractor's effect on the decision variable, $\beta_{2,h} d_s + \beta_{3,h}$, is the same in every session at a given amplitude; the shift in µm also depends on the session's slope and lapse rates. A *typical session* sets all z's to zero. In a typical session the unimanual PSE differs between the 32 and the 6 µm sessions by $-26\,\lambda_h / \mu_{1,h}$ µm. Averages "over sessions" are means over the 42 sessions (or over the sessions with a given distractor amplitude), computed for each posterior draw. All intervals are 95% highest-density intervals: for PSE and JND over 2000 draws, for parameters and lapse rates over all 20,000.

Table 2. Model E's hand-level parameters: posterior mean and 95% highest-density interval (20,000 draws). Logit scale unless noted; d_s is in units of σ_a.

Parameter	Meaning	Left hand	Right hand
$\mu_{0,h}$	mean session bias at $d_s = \bar{d}$	−1.40 [−1.65, −1.13]	−1.02 [−1.29, −0.74]
$\sigma_{0,h}$	session-to-session sd of the bias	0.62 [0.41, 0.83]	0.69 [0.50, 0.91]
λ_h	criterion trend per unit of d_s	−0.53 [−0.88, −0.17]	−1.03 [−1.42, −0.66]
$\mu_{1,h}$	mean slope	4.64 [4.25, 5.04]	4.39 [4.05, 4.73]
$\sigma_{1,h}$	session-to-session sd of the slope	0.55 [0.02, 0.98]	0.35 [0.00, 0.68]
$\beta_{2,h}$	distractor amplitude term	0.72 [0.36, 1.08]	0.66 [0.34, 1.01]
$\beta_{3,h}$	distractor presence term	0.15 [−0.40, 0.66]	0.76 [0.24, 1.29]
μ_h^{hi}	mean unimanual "high" lapse logit	−5.46 [−7.10, −3.91]	−5.04 [−6.60, −3.66]
σ_h^{hi}	its session-to-session sd	0.35 [0.00, 0.87]	0.36 [0.00, 0.88]
μ_h^{lo}	mean unimanual "low" lapse logit	−3.97 [−5.62, −2.55]	−3.97 [−5.55, −2.62]
σ_h^{lo}	its session-to-session sd	0.37 [0.00, 0.91]	0.37 [0.00, 0.92]
$\mu_h^{\mathrm{hi,bi}}$	mean bimanual "high" lapse logit at $d_s = \bar{d}$	−2.50 [−3.42, −1.67]	−1.84 [−2.52, −1.17]
$\sigma_h^{\mathrm{hi,bi}}$	its session-to-session sd	1.02 [0.47, 1.61]	0.45 [0.00, 0.98]
κ_h^{hi}	change per unit of d_s	2.51 [1.15, 3.91]	0.64 [−0.27, 1.66]
$\mu_h^{\mathrm{lo,bi}}$	mean bimanual "low" lapse logit at $d_s = \bar{d}$	−3.04 [−4.01, −2.19]	−3.40 [−4.38, −2.51]
$\sigma_h^{\mathrm{lo,bi}}$	its session-to-session sd	0.57 [0.00, 1.24]	0.39 [0.00, 0.94]
κ_h^{lo}	change per unit of d_s	−0.69 [−2.09, 0.73]	−0.65 [−1.99, 0.61]

3 The other models, as changes to E

The other tied models.

- **E+shared**: $\beta_{0,hs}$ gains a daily offset common to both hands, $+\sigma_v v_s$ with $v_s \sim \mathcal{N}(0,1)$ and $\sigma_v \sim \mathrm{HalfNormal}(1)$.
- **D** (= C+LKP, previous recommendation): no trend ($\lambda_h = 0$), and the distractor terms vary between sessions, $\beta_{2,hs} = \mu_{2,h} + \sigma_{2,h} z_{hs}^{(2)}$ and $\beta_{3,hs} = \mu_{3,h} + \sigma_{3,h} z_{hs}^{(3)}$, with $\mu_{2,h}, \mu_{3,h} \sim \mathcal{N}(0,1)$ and $\sigma_{2,h}, \sigma_{3,h} \sim \mathrm{HalfNormal}(1)$.
- **D+crit**: D with the trend term $\lambda_h(d_s - \bar{d})$ of E.
- **D+order**: D with a linear trend in time instead, $\tau_h t_s$ with $t_s = (s - 21.5)/10$ for $s = 1, \ldots, 42$ and $\tau_h \sim \mathcal{N}(0,1)$.
- **D+crit+order**: D with both terms.

Other models referred to above.

- **D4**: E without the trend ($\lambda_h = 0$). **D+shared**: D with the daily offset of E+shared.
- **D3+crit**: E with the slope and all lapse rates fixed across sessions ($\sigma_{1,h} = \sigma_h^k = \sigma_h^{k,\mathrm{bi}} = 0$; κ_h^k kept), so only $\beta_{0,hs}$ varies between sessions.
- **C** (Vera): $\eta_i = \beta_{0,hs} + \beta_{1,hs} x_i + \beta_{2,hs} d_i$ with $\beta_{2,hs}$ as in D, no β_3, no trend, and the same lapse rates $\frac{1}{4}\,\mathrm{logistic}(\omega_{hs}^k)$ on unimanual and bimanual trials.
- **C+L**: C with bimanual lapse logits of their own, $\omega_{hs}^{k,\mathrm{bi}} = \mu_h^{k,\mathrm{bi}} + \sigma_h^{k,\mathrm{bi}} z_{hs}^{(k,\mathrm{bi})}$ (priors as in E). **C+LP**: D without κ ($\kappa_h^k = 0$).
- **D + bimanual slope** (C+LKPS): η_i gains $\delta_{hs}\, c_i\, x_i$ with $\delta_{hs} = \mu_{\delta,h} + \sigma_{\delta,h} z_{hs}^{(\delta)}$, $\mu_{\delta,h} \sim \mathcal{N}(0,1)$, $\sigma_{\delta,h} \sim \mathrm{HalfNormal}(1)$.
- **Capture models** (C+LPMdist, C+LPMmax): C+LP, with the probability of "high" on a bimanual trial replaced by $(1-\pi_h)\, p_i + \pi_h\, p_i^{\mathrm{cap}}$. Here $p_i^{\mathrm{cap}} = \gamma_u^{\mathrm{hi}} + (1 - \gamma_u^{\mathrm{hi}} - \gamma_u^{\mathrm{lo}})\,\mathrm{logistic}(\beta_{0,hs} + \beta_{1,hs} x_i^{\mathrm{cap}})$ is the unimanual curve (γ_u: the unimanual lapse rates), evaluated at the distractor's position on the stimulus scale, $x_i^{\mathrm{cap}} = d_i - 28/\sigma_a$ (Mdist), or at $\max(x_i,\, d_i - 28/\sigma_a)$ (Mmax); $\mathrm{logit}\,\pi_h \sim \mathcal{N}(-2.5, 1.5^2)$.
- **B** (Vera): four groups $g = (h, c)$, each with its own hierarchy: $\eta_i = \beta_{0,gs} + \beta_{1,gs} x_i$ and lapse rates $\frac{1}{4}\,\mathrm{logistic}(\omega_{gs}^k)$, with $\beta_{0,gs}, \beta_{1,gs}, \omega_{gs}^k$ formed and given priors as $\beta_{0,hs}, \beta_{1,hs}, \omega_{hs}^k$ in E (without λ). B has no distractor covariate: each session's bimanual curve has its own bias, slope and lapse rates.

4 How the models were scored

Each model was fitted with NUTS (4 chains, 5000 draws after 1000 tuning steps, target acceptance 0.9), giving 20,000 posterior draws. With ϕ the hand-level parameters, θ_s all session-s parameters of both hands, $\mathbf{y}_s$ the responses of session s and $\mathbf{y}_{-s}$ all other responses:

$$\mathrm{elpd}_{\mathrm{trial}} = \sum_{i=1}^{N} \log p(y_i \mid y_{-i}), \qquad \mathrm{elpd}_{\mathrm{session}} = \sum_{s=1}^{S} \log p(\mathbf{y}_s \mid \mathbf{y}_{-s}),$$

$$p(\mathbf{y}_s \mid \mathbf{y}_{-s}) = \int p(\mathbf{y}_s \mid \phi)\, p(\phi \mid \mathbf{y}_{-s})\, d\phi, \qquad p(\mathbf{y}_s \mid \phi) = \int p(\mathbf{y}_s \mid \theta_s, \phi)\, p(\theta_s \mid \phi)\, d\theta_s.$$

The trial-level score is estimated by Pareto-smoothed importance sampling (PSIS-LOO) over the 20,000 draws, and $p_{\mathrm{loo}} = \sum_i \log\overline{\big(p(y_i \mid \theta)\big)} - \mathrm{elpd}_{\mathrm{trial}}$, where the bar is the average over posterior draws. For the session score, $p(\mathbf{y}_s \mid \phi)$ is computed for each of 1000 evenly thinned posterior draws of ϕ by importance sampling with 2000 samples (proposal: 90% a t_5 distribution from a Gaussian approximation of $p(\mathbf{y}_s \mid \theta_s, \phi)\, p(\theta_s \mid \phi)$, its scale widened 1.2 times, and 10% $p(\theta_s \mid \phi)$ itself). The outer integral uses $p(\phi \mid \mathbf{y}_{-s}) \propto p(\phi \mid \mathbf{y}) / p(\mathbf{y}_s \mid \phi)$ and is computed by PSIS over the same 1000 draws. For two models, with δ_s the difference of their scores summed over the trials of session s (or the difference of their session scores),

$$\Delta = \sum_{s=1}^{S} \delta_s, \qquad \mathrm{SE}(\Delta) = \sqrt{S \cdot \frac{1}{S-1} \sum_{s=1}^{S} \big(\delta_s - \bar{\delta}\big)^2},$$

and a difference is treated as clear when $|\Delta| > 2\ \mathrm{SE}(\Delta)$.